Topic 8: Network Security, Cryptography & Emerging Technologies

Course learning outcome 7 (network security & algorithms), presentation-layer cryptography (weeks 8-9) and week 12 emerging technologies.

How often this topic is examined

Question asked	Paper(s)	Marks
Four network security threats	SE2024 Q2d	3
Harm caused by Trojans, worms, ransomware, spam	SE2024 Q5a	4
DES vs AES	2023 Q4c	4
Key Distribution Centre; Certification Authority	2023 Q4d	6
How a VPN works	ISA2024 Q2a	3
Emerging technologies (week 12)	Course outline	possible

1. Security goals (use in any security answer)

- **Confidentiality** - only authorised people can read the data (encryption).
- **Integrity** - data is not altered in transit (hashes, MACs, digital signatures).
- **Availability** - services are available when needed (protection against DoS).
- Plus **authentication** (proving identity) and **non-repudiation** (sender can't deny sending).

2. Network security threats

> **EXAM QUESTION** SE2024 Q2d **[3]**
> Discuss any four network security threats you are familiar with.

Threat	Explanation
Malware	Malicious software - viruses, worms, Trojans, ransomware, spyware - that damages systems or steals data
Phishing / social engineering	Fake emails/websites/calls trick users into revealing passwords or card details or installing malware
Denial of Service (DoS/DDoS)	Flooding a server/network with traffic (e.g. SYN flood, botnet) so legitimate users cannot access it - attacks availability
Man-in-the-middle / eavesdropping (sniffing)	Attacker intercepts and possibly alters traffic between two parties (ARP spoofing, rogue Wi-Fi) - attacks confidentiality & integrity
Spoofing	Faking an IP/MAC/email address or DNS response to impersonate a trusted host
Password attacks / unauthorised access	Brute-force, dictionary attacks, stolen credentials
SQL injection / insider threats / zero-day exploits	Exploiting application flaws or trusted employees misusing access

Made by King Tine, the AI Engineer

EXAM QUESTION SE2024 Q5a **[4 x 1]**
Explain the harm caused by: Trojans, Worms, Ransomware, Spam.

- **Trojan (horse)** - malware disguised as legitimate software; once installed it opens a **backdoor**, giving attackers remote control, stealing data (passwords, banking details) or downloading more malware. It does **not self-replicate**.
- **Worm** - **self-replicating** malware that spreads across networks **without user action** by exploiting vulnerabilities; it consumes bandwidth and system resources (slowing/crashing networks) and can carry payloads (e.g. WannaCry, Conficker).
- **Ransomware** - **encrypts the victim's files** (or locks the system) and **demands a ransom** (usually cryptocurrency) for the decryption key; causes data loss, downtime and financial loss.
- **Spam** - unsolicited bulk email; wastes bandwidth, storage and time, reduces productivity, and is a common carrier of **phishing** links and malware.

3. Cryptography basics

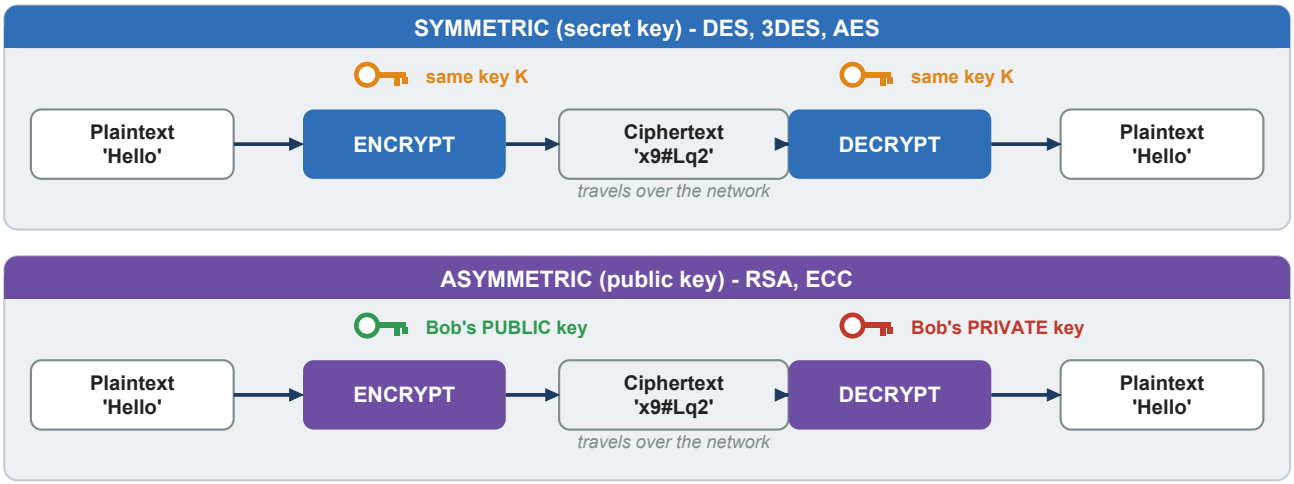


Figure: symmetric vs asymmetric encryption

EXAM QUESTION 2023 Q4c **[4]**
Highlight the major differences between DES and AES encryption methods.

Feature	DES (Data Encryption Standard)	AES (Advanced Encryption Standard)
Adopted	1977 by NIST (from IBM's Lucifer)	2001 by NIST (Rijndael algorithm) - replaced DES
Key length	**56 bits** (64 with parity bits)	**128, 192 or 256 bits**
Block size	**64-bit** blocks	**128-bit** blocks
Structure	**Feistel network**	**Substitution-permutation network** (SubBytes, ShiftRows, MixColumns, AddRoundKey)
Rounds	**16** rounds	**10 / 12 / 14** rounds (for 128/192/256-bit keys)
Security	**Insecure** - broken by brute force (1998-99, under 24 h); 3DES used as stop-gap	**Secure** - no practical attack; used worldwide (Wi-Fi WPA2/3, HTTPS, disk encryption)
Speed	Slower in software	Faster and efficient in hardware and software

EXAM QUESTION 2023 Q4d **[3 + 3]**
Define the following terms as applied in network security: i) Key distribution center ii) Certification authority.

i) Key Distribution Centre (KDC): a **trusted third party** in **symmetric-key** systems that shares a secret key with every user and **creates and distributes temporary session keys** so two users can communicate securely without having met before. Instead of n(n-1)/2 keys, each user only needs one key with the KDC. Example: **Kerberos** (Authentication Server + Ticket-Granting Server issue tickets). Steps: Alice asks the KDC for a session key with Bob -> KDC sends the session key encrypted with Alice's key, plus a 'ticket' (the session key encrypted with Bob's key) -> Alice forwards the ticket to Bob.

ii) Certification Authority (CA): a **trusted organisation that issues and signs digital certificates** binding a **public key to an identity** (person, server, domain) in a **Public Key Infrastructure (PKI)**. The CA verifies the identity, then signs the certificate (X.509) with its own private key so anyone can verify it with the CA's public key. It also **revokes** compromised certificates (CRL/OCSP). Examples: DigiCert, Let's Encrypt, GlobalSign - used for HTTPS/TLS.

4. VPN

EXAM QUESTION ISA2024 Q2a **[3]**
VPN can be essential in network security, briefly explain how VPN works.

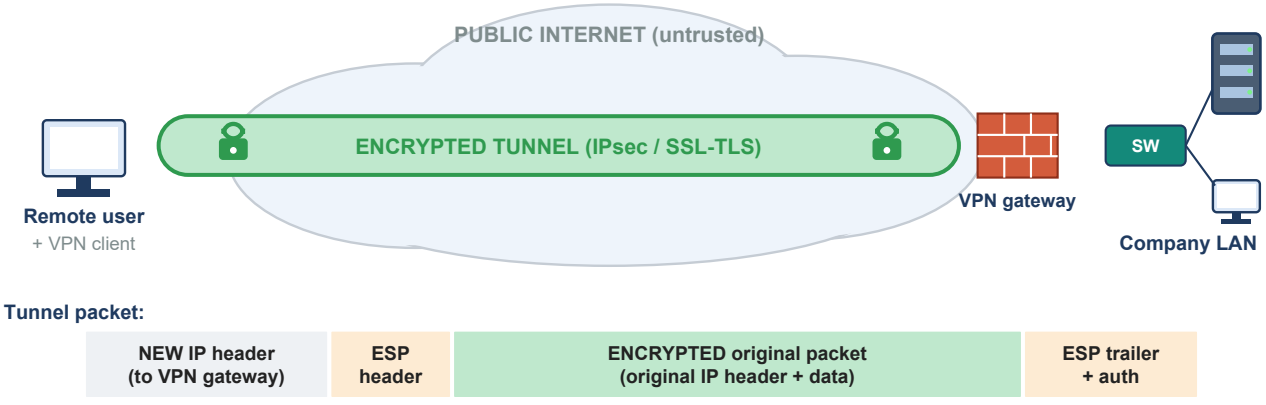


Figure: how a VPN tunnel works

- A **Virtual Private Network** creates a **secure, encrypted 'tunnel'** across a public network (the Internet) so remote users/branches can use the private network as if they were directly connected.
- The VPN client and VPN server/gateway **authenticate** each other (passwords, certificates, pre-shared keys) and agree encryption keys.
- Each original packet is **encrypted and encapsulated** (tunnelled) inside a new packet addressed to the VPN gateway; the gateway decrypts it and forwards it into the LAN.
- Provides **confidentiality** (encryption), **integrity** (hashing) and **authentication**; hides the user's real IP.
- Protocols: **IPsec** (AH/ESP, site-to-site), **SSL/TLS VPN**, **OpenVPN**, **WireGuard**, L2TP/IPsec. Types: **remote-access** and **site-to-site**.

5. Other security controls (good extra points)

- **Firewall** - filters traffic between networks using rules (packet filtering, stateful inspection, application/next-gen firewalls).

- **IDS/IPS** - detects/prevents intrusions by signature or anomaly.
- **DMZ** - separate zone for public servers (web, mail) between Internet and internal LAN.
- **Access control & authentication** - strong passwords, multi-factor authentication, 802.1X, ACLs on routers.
- **Encryption** - HTTPS/TLS, WPA3 for Wi-Fi, SSH instead of Telnet.
- **Patching, antivirus, backups, user awareness training, network segmentation/VLANs.**
- **Hashing / digital signatures** - MD5, SHA-256 for integrity; signature = hash encrypted with sender's private key (non-repudiation).

6. Emerging technologies (week 12 - 'discussion of modern trends')

Technology	What it is / why it matters
5G	Mobile networks with very high speed (up to ~10 Gbps), ultra-low latency (~1 ms) and massive device density - enables IoT, smart cities, remote surgery
Internet of Things (IoT)	Billions of sensors/devices connected to the Internet (smart meters, agriculture sensors) - needs IPv6, low-power protocols (Zigbee, LoRaWAN), raises security issues
Cloud computing	On-demand computing & storage over the Internet: IaaS, PaaS, SaaS (AWS, Azure, Google Cloud)
SDN & NFV	Software-defined, programmable networks; Network Function Virtualisation runs routers/firewalls as software on standard servers
Edge / fog computing	Processing data near where it is generated to cut latency and bandwidth
Wi-Fi 6/6E/7	802.11ax/be: higher throughput, OFDMA, better performance in crowded areas
Satellite Internet (LEO)	Low-Earth-orbit constellations e.g. Starlink - broadband in rural areas (now in Zimbabwe)
AI in networking, Zero Trust, IPv6 adoption, blockchain	AI-driven network monitoring/automation; 'never trust, always verify' security model

REMEMBER: Trojan = disguised, backdoor, no self-copy. Worm = self-copies over the network. Ransomware = encrypts and demands payment. DES 56-bit key/64-bit block/16 rounds - broken; AES 128/192/256-bit key/128-bit block/10-14 rounds. KDC = symmetric session keys (Kerberos). CA = signs certificates for public keys (PKI). VPN = encrypted tunnel over the Internet.